Chapter # 8

CHROMOSOME and DNA

- **Chromosomes:- (Chroma= coloured, soma= body)**

Discovered by walther fleming in 1882 from dividing cells of salamander larva.
He used Perkins Aniline dye for staining.
The chromosomes can be described as thread like structures present in the nucleus, bearing hereditary characters in the form of genes, present in pairs and their number remains constant from generation to generation.

STRUCTURE:

Each chromosome consists of two very fine threads like structure called **chromatid.** Chromatids are joined together at a point called **centromere OR Primary constriction**.

SISTER CHROMATIDS:

"The two chromatids of the same chromosomes are called **sister** chromatids".

NON-SISTER CHROMATIDS:

"The two chromatids of two different chromosomes are called **non-sister**chromatids.

Sometimes another constriction is also present called **secondary constriction** or **nuclear organizer** which developed from nucleolus during interphase. Due to secondary constriction the end of chromosome becomes knob like called **satellite** and the terminal part of chromosome is called **telomeres** which contain junk DNA. It prevent the attachment of two chromosomes.

CHEMICAL COMPOSITION OF CHROMOSOMES

Chemically chromosomes are composed of nucleoproteins that is 40% DNA and 60% proteins. RNA is also present but it is not the constituent part of chromosome.

Chromosome contain highly condensed double stranded very long , unbroken DNA through the entire length of chromosome.If the single set of chromosomes from human cell are laid in a straight line, it would be more than seven feet or 2 meter long which is too long to be fit in a cell.

HOW COILING OCCURS

This organization of chromosome occurs in three stages

(1) NUCLEOSOME STRING FORMATION

The chromosome contain positively charged histone proteins which are basic in nature due to presence of arginine and lysine amino acids, while the DNA strand is negatively charged due to presence of phosphate. Every 200 nucleotides, the DNA duplex is coiled around a core of eight histone protein (octamer) forming a complex called **nucleosome.** Each nucleosome is linked with each other by a 2nm thick DNA called a linker DNA, In this way string of nucleosome is formed which is 10 nm thick.

(2) FORMATION OF CHROMATIN FIBER

This nucleosome acts as magnet and begins to coil at its own axis results in the formation of 30 nm thick fibre called chromatin fiber.

This chromatin fiber consist of two regions

(a) HETEROCHROMATIN

"Highly condensed portions of the chromatin are called heterochromatin".

They are dark in color having double amount of DNA. These are inactive and do not take part in the formation of RNA. The gene in this region never are expressed.

(b) EUCHROMATIN REGION

"Portion of chromatin which is not condensed (except during cell division) is called euchromatin".

These regions have light color. They form larger part of chromosome. These are active regions and form RNA. The genes in this region can be expressed.

(3) SUPERCOIL FORMATION TO FORM CHROMATIDS

The chromatin fiber starts higher order of coiling to form a super coiled structure of 300 nm which immediately develops into 700 nm thick chromatids.

GENE AND GENE LOCUS

GENE

In 1860 Gregor John mendel first introduced the concept of **factor**. Which controls the expression of a character and represented it by letter.

In 1906 William batson and Edward strasberger used the term **pengene** for the gene.

In 1909 William Johnson introduced the term **gene.**

The gene is composed of DNA which consists of many nucleotide in a sequence and has the information to synthesize a specific polypeptide chain (protein) which performs a specific reaction in the living Organism.

ALLELES

The alternative form of a gene for a trait is called allele. Mendel found two alleles for each character. He observed seven characters in Pea plant and found two alternative forms in each character. For example the length of Pea plant had two characters tall and dark, flower colour had two alternative forms like white flowered pea plant and purple flowered pea plant etc.

GENE LOCUS

The position of gene on the chromosome is called locus. A diploid cell contain two genes for each character and the haploid cell (gametes) contain one gene for each character.

HOMOLOGOUS CHROMOSOMES

If the pair of chromosome have morphologically similar chromosomes then this pair is known as homologous chromosomes

HETEROLOGOUS CHROMOSOMES

The pair of chromosome in which both the members are morphologically different.

HISTORY OF CHROMOSOMAL THEORY OF INHERITANCE

- In 1860s Mendel presented his laws of inheritance but his work was not recognized at that time.
- Mendel's work was recognized by three scientists Hugo de Varies, Carl correns and Eric Von in 1900.
- Carl corren rediscovered mendel's work independently in 1890 and published his paper on 25th January 1900 with the title **Mendel's Law Concerning The Behaviour Of Progeny Of Racial Hybrids.**
- Walther Fleming, Waldeyer, Walter Sutton ,TH Morgan and Theodore bovan worked on structure and no of chromosomes.
- Walter Fleming discovered chromosomes in 1882 from the dividing cells of salamander larvae
- Walter Sutton and Theodore Boven discovered that diploid cells contain chromosomes in pairs while haploid cells i.e egg and sperm contain only one member of each pair of chromosome
- Walter Sutton observed the cells of grasshopper and found that segregation pattern of chromosomes during meiosis resembles the segregation pattern of mendel's gene. He published his work in a paper with the title **the chromosomes in heredity.**

STATEMENT AND EVIDENCES OF CHROMOSOMAL THEORY OF INHERITANCE
OR

IMPORTANT POSTULATES OF CHROMOSOMAL THEORY OF INHERITANCE

It was first formulated by Walter Sutton in 1902

(1) Sexual reproduction involves the initial union of egg and sperm ,if mendel's model is correct then these two gametes must make equal hereditary contribution, however sperms contain little cytoplasm. Therefore it is considered that hereditary material must be present within the nucleus of gametes.

(2) Gametes have one copy of each pair of homologous chromosomes while deploid individuals have two copies. It is consistent with mendel's model in which deploid individuals have two copies of each gene and gametes have one.
(3) During meiosis each pair of homologous chromosome orients on the metaphase plate independent of any other pair and assort into different gametes as explained by Mandel's law of independent assortment of factors.
(4) The chromosomes segregate during meiosis in a manner similar to the mendelian factors segregate according to law of segregation

OBJECTION ON THE CHROMOSOMAL THEORY OF HEREDITY

If independent assortment of mendeliantraits reflect the independent assortment of chromosomes in meiosis then why does the number of characters or factors in a given kind of Organism often exceeds then the number of chromosome pairs the Organism posses?

EXPERIMENTAL VERIFICATION OF CHROMOSOMAL THEORY OF INHERITANCE

In 1910 Thomas hunt Morgan studied Drosophila melanogaster (fruit fly). The normal eye color is red and he found male drosophila fly with white eyes (mutant).
He crossed **white eyed male** with **red eyed female** and in **F1 all the flies were with red eyes** then **he crossed F1 flies with each other** that is red eyed female and red eyed male which produced 4252 flies out of which 782 (18%)were white eyed. This ratio of red eye to white eyes was greater than 3:1.
This result did not provide evidence that eye color segregates but there was something strange and totally unpredicted by Mendel that is all the white eyed F2 flies were male.
How could this result be explained? Perhaps it was impossible for white eyed female flies to exist?
The Morgan tested this by crossing red eyed female of F1 with white eyed male
He obtained red eyed, white eyed male and female in the ratio of 1:1:1:1.
As Mendel predicted that genes segregate.
Then why were there no white eyed female among the progeny of original cross?
The answer is that genes for eye color in drosophila resides on only X chromosome and absent from y chromosome knowing that white eye trait is recessive to red eye trait, Morgan's result was a natural consequence of mendelian assortment of chromosome.
This experiment presented first clear evidence that genes reside on the chromosomes.

HEREDITARY MATERIAL

INTRODUCTION:

In the first half of 20^{th} century, It was confirmed by chromosomal theory that genes are located on the chromosomes.investigation showed that chromosomes and genes are composed of DNA and protein. The question arises that which molecule function as hereditary material either protein or DNA.

EVIDENCE OF DNA AS HEREDITARY MATERIAL

1- GRIFFITH EXPERIMENT:

In 1928 Griffith isolated two strains of bacterium, one was smooth surface 'S-type' capsulated strain and other was rough surface 'R-type' non-capsulated strain. S-type strain was virulent while R-type strain was non-virulent. By injecting these strains into mice, he observed following results.

1- Laboratory mice were injected with live R-type strain, no illness effects were observed.
2- Mice were injected with live S-type strain; the mice contracted pneumonia and died.
3- S-type strain were killed with high temperature and then injected into mice, no illness effects were observed.
4- Live R-type strain was mixed with heat killed S-type strain and injected to mice than highly mortality of mice was observed.

CONCLUSION:

He had no any satisfactory explanation for 4th experiment, It was thought that S-type somehow became alive but it was not acceptable.

2-AVERY, MACLEOD & MCCARTY'S STUDIES:

In 1944 Avery, Macleod and McCarty discovered the true identity of the transforming substance.

STUDY:

They tested the heat-killed bacteria for their transforming ability and found out that it is not RNA or protein but only DNA that possessed the transforming ability. If the enzyme Deoxyribo-nuclease that destroys the DNA was added to the bacteria, the transforming ability was lost.

CONCLUSION: This study proved beyond any doubt that DNA must be the genetic material.

3-HERSHEY AND CHASE EXPERIMENT:

In 1952, Alfred Hershey and Martha chase set out an experiment to see that which material bacteriophage work as hereditary material Either DNA or protein.

EXPERIMENT:

For this purpose, they labeled bacteriophage DNA with radioactive isotope of phosphorus (p^{32}) and protein with radioactive sulphur (S^{35}). The labeled bacteriophage was permitted to attack on bacteria. After some time, the new phages formed contained only p^{32}indicating the presence of DNA molecules. However the S^{35} was present in the medium.

CONCLUSION:

This experiment proved that when phage infects the bacteria, only phage DNA enters the cells and the phage protein remains outside. This experiment provided good evidence that DNA is the hereditary material which is transmitted from one generation to the next

DNA replication

The process of producing exact copy of DNA molecule using old DNA strand as template with the help of enzymes is called DNA replication.

This process occurs during S phase of interphase. In this process the molecule of DNA which is replicated is called parental DNA while the molecule of DNA produced is called daughter DNA.

Models of DNA replication

There are three different models proposed to explain the replication of DNA that is
(i) conservative
(ii) Dispersive model
(iii) Semi Conservative and

(i) Conservative model
This model states that after replication of DNA, two daughter DNA molecules are formed. One daughter DNA molecule contains both parental strands while the other daughter DNA molecule contains newly synthesized DNA strands. It means the parental DNA remains intact.

(ii) Dispersive model
According to this model during DNA replication, the parental DNA molecule breaks into its nucleotides then these nucleotides combine together with new nucleotides to form two new daughter DNA molecule which have mixture of old and new DNA.

(iii) Semi conservative model
This model states that the two parental DNA strands separate and each of these strand serve as a template (model DNA) for synthesis of new DNA strand. As a result two DNA double Helix are formed both of which contain one parental DNA strand and one newly synthesized DNA strand.

SEMI CONSERVATIVE REPLICATION OF DNA

The form of DNA replication suggested by the Watson – Crick Model is called semi conservative.

MATTHEW MESELSON & FRANKLIN SATHL' EXPERIMENT:

INTRODUCTION:

They grew bacteria (several generations) in a medium containing heavy isotope of Nitrogen N^{15}. They observed that DNA of bacteria was denser than the normal due to heavy isotope of Nitrogen. Then they transferred these bacteria to the medium containing lighter isotope of Nitrogen N^{14}.this time density of DNA is decreased to intermediate level because newly synthesized DNA has one helix of N^{15} and other of N^{14}. Another round of replication takes place in the same lighter isotope medium. In the result, one intermediate (N^{15} and N^{14}) and other light density DNA molecules are formed.

CONCLUSION:

By these observations, they concluded following results;

- In the first round of replication, each daughter DNA duplex possessed one heavy helix and one light helix.
- When this duplex replicated, heavy strands to form heavy duplex and light strands to form a light duplex.

This experiment shows that "DNA replicates in a semi conservative manner".

Process of DNA replication

It involves many enzymes and proteins.
The basic steps of replication in prokaryotes and eukaryotes are similar with minor differences.

(a) Origin of replication

The replication of DNA begins at a specific site called origin of replication site which consists of a specific sequence of nucleotides. In a prokaryotic cell there is only one origin of replication site but in the eukaryotic DNA there may be more than one origin of replication sites.

The DNA helicase enzyme and DNA gyrase enzyme work together at origin of replication site.

The DNA helicase enzyme breaks the hydrogen bond present between complementary base pairs of DNA duplex and the DNA gyrase enzyme facilitates the DNA helicase enzyme in unwinding of DNA duplex, as a result complementary strands of DNA duplex gradually separates from each other and forms a bubble like structure called replication bubble. Each side of replication bubble is termed as replication fork . A protein named single stranded binding protein (SSB) prevent the rebinding of complementary strands at replication bubble.

(b) Elongation of new strands

The step during which daughter DNA strands are formed using parental DNA strands as template strand. This process is catalyzed by three types of DNA polymerase enzymes.

This elongation of new strand takes place from 3 end of replication fork, thus the direction of replication is 5 end to 3 end, Along with the template strand.

Leading strand

The daughter DNA strand which elongates towards the direction of replication fork and it is a continuous DNA strand.

Lagging strand

The new strand of DNA which is formed away from the direction of replication fork and is formed in short segments which are called okazaki fragments. Each okazaki fragment contains 100 to 200 nucleotides. In a eukaryotic cell these fragments are joined together by an enzyme called DNA ligase to form a news trend. This strand is known as lagging strand.

(c) Priming DNA synthesis

Primer is a short segment of RNA containing 10 nucleotides. In a Eukaryotic cell, this primer is formed by the activity of primase Enzyme.

The primer functions as binding site for DNA polymerase enzyme to initiate the replication of DNA, because DNA polymerase enzyme can add nucleotides to an existing nucleotide chain only it cannot start from the scratch.

(d) Types of DNA polymerase enzymes and their functions

There are three types of DNA polymerase enzymes which perform different roles

(i) DNA polymerase I

It is responsible for replacement of RNA primer by DNA nucleotide during termination.

(ii) DNA polymerase II

It performs the proof reading of newly formed DNA strand and repairs the DNA damages

(iii) DNA polymerase III

It is the main enzyme of the replication which performs elongation and formation of new daughter strand.

(e) Termination

In this phase of DNA replication, following processes are performed

- The DNA polymerase I replaces the RNA primer with the DNA nucleotides.

- The DNA polymerase II perform proof reading and removes any wrong nucleotide if added mistakenly in the newly synthesized DNA strand.
- The okazaki fragments in the lagging strand are joined together by DNA ligase enzyme to form a continuous DNA strand.

Replication of DNA as process of stability and variability

Stability

During replication of DNA, There are chances of mistakes due to speed of replication that is 50 to 500 nucleotide per second are added or DNA polymerase III occasionally attaches incorrectly matched bases (one mistake for every 10,000 base pairs). In a mammalian cell rarely a mistake may occur like one mistake for every billion base pairs.

However, The stability and consistency of DNA is maintained generation after Generation Due to hydrogen bonding present between the base pairs of old and newly formed DNA strand, presence of proof reading system in which DNA polymerase II replaces mismatched base pairs.

Variability

The variation in DNA can occur due to mutation and Tautomeric forms.

(i) Mutation

The permanent change in DNA sequence that could not be corrected by DNA polymerase II enzyme during proof reading. Hence an altered DNA is transferred into the next generation.

(ii) Tautomeric forms OR Shift

The nitrogenous bases exists in different chemical forms like tautomers i.e in which H group can shift from one atom (position) another atom (position).

Most of the variations in DNA occurs due to attachment of tautomers in DNA which is a mismatch base pairing results in variability.

Gene expression

The process in which the information present on gene is used to produce a protein is called gene expression.

Central dogma of gene expression

It means that How our genetic information flows from DNA to RNA for synthesizing a protein which work as an enzyme. This genetic information from DNA to protein flows in two steps Transcription and translation.

(1) Transcription

The process in which DNA produces mRNA which carries information of DNA in coded form from nucleus to the ribosomes present in the cytoplasm, to produce a particular protein.

(2) Translation

The process in which the information present on mRNA is converted into correct sequence of amino acids with the help of tRNA and rRNA.

GENETIC CODE

“Genetic code is a combination of three nucleotides which has genetical information to form a particular amino acid”.

The important work done by nucleotides present in DNA. Like DNA, RNA also has four nucleotides.

1- Adenine 2- Guanine 3- Cytosine 4- Uracil

The genetic codes use three bases to specify each amino acid:

There are 20 different amino acids that commonly take part in protein synthesis. To understand this mechanism, we suppose that one nucleotide codon for one amino acid, so in this way only 4 amino acids can be formed and 16 amino acids were left without coding. If we take 2 nucleotides code then there are only 16 codes possible. This problem solves when we take the sequence of three nucleotides code for each amino acid. In this way, we obtained 64 codes. These codes are greater in numbers that of amino acids, so more than one code can be coded a single amino acid. Out of 64 codons, three codons (UAG, UAA & UGA) are used as stop signals and they do not code for any amino acid whereas AUG is considered as start codon a and also codes for methionine.

Genetic code gives three information to cell

(i) The amino acids required for the protein.

(ii) The sequence of these amino acids in the protein.

(iii) the length of polypeptide chain.

Mechanism of transcription

The process in which particular part of DNA is copied in the mRNA is called transcription. It has two limitations.

(i) Only selected part of gene of DNA is prescribed. for example cells of hair follicle transcribe only information about keratin protein.

(ii) Secondly it occurs only when it is required and it copies only one strand of DNA.

The DNA strand which is copied to produce mRNA is called template strand or antisense strand while the opposite strand is termed as coding strand or sense strand.

Transcription is a continuous process however, we can divide it into three phases initiation, elongation and termination.

(a)Initiation

Transcription begins with the binding of RNA polymerase enzyme at the promoter region that comprises of binding site for attachment of RNA polymerase enzyme. In prokaryote cell these binding sites are TATAAT also called -10 sequence and TTGACA also called -35 sequence whereas in eukaryotic cell TATA (tata box) also called -25 sequence and CAAT (CAAT box) also called -70 sequence. Names of this sequence refer to their position in promoter region.

(b) Elongation

As the RNA polymerase binds to the promoter, it unzip DNA double Helix and Forms a transcription bubble. RNA polymerase begins to arrange the free RNA nucleotides present in the nucleus to make and elongate RNA strand in 5 to 3 direction complementary to the template or antisense DNA strand. The base pairing rule is same for transcription as for DNA replication except that uracil is paired with adenine instead of thymine. This event continues till the RNA polymerase reaches the terminator region of the gene.

(c) Termination

The termination region is a sequence of DNA bases which consists of series of GC base pairs followed by AT base pairs. The part of mRNA which is transcribed in this region projects to form a loop like structure called GC hairpin, followed by a small tail of poly U nucleotide. The GC hairpin causes the RNA polymerase to stop synthesis of RNA.

As the RNA polymerase reaches, the Terminator Region it causes the RNA molecule to separate form DNA strand and RNA polymerase enzyme secondly the RNA polymerase detached from template strand of DNA.

Modification in mRNA before translation

In prokaryotic cell transcription and translation takes place in cytoplasm therefore it does not require any modification in mRNA. Whereas in a eukaryotic cell the mRNA is modified before translation Because in eukaryotic cell, the transcription occurs in nucleus while translation occurs in cytoplasm. So mRNA has to move from nucleus to cytoplasm and during this journey enzymes like phosphatases and nucleases may degrade it, furthermore the eukaryotic mRNA contains some non coding regions called introns which are to be removed.

The removal of introns and maturation of primary RNA to secondary RNA is called RNA splicing. This takes place in three steps.

(1) A group of small RNPS(ribonucleotide proteins) binds to introns of primary mRNA.

(2) Binding of small RNPS brings the 5 end and 3 ends of intron closer which makes a loop, in this way the ends of Exon also come close and ultimately join together.

(3) The Introns removed and splice site connect to make mRNA then small RNPs detach from intron and reused.

Secondly a cap in the form of 7 methyl GTP is attached at 5 end of mRNA and a polyp A tail is attached at 3 end of mRNA. These two modifications prevent mRNA to be degraded by phosphatase and nuclease enzymes.

Translation

In translation, mRNA produced by transcription is decoded by the ribosomes to produce a specific amino acid chain or polypeptide or protein. This process can be divided into 4 phases

(i) Activation of amino acids

Activation of amino acid refers to the binding of free amino acids to the 3 end of particular tRNAmolecule. This binding is catalyzed by aminoacyl synthase enzyme.

The amino acid attaches with 3 end of tRNA called amino acid site.

(ii) Formation of initiation complex

The process of translation actually begins with the Formation of initiation complex. It is formed by the combination of ribosomal subunit, mRNA and first aminoacyl tRNA complex.

A tRNA molecule carrying chemically modified methionine called N formyl methionine binds smaller ribosomal subunit. This binding is catalyzed by initiation factor I. At the same time 5end of mRNA molecule binds to the smaller subunits of ribosomes with the help of another enzyme called initiation factor II, initiation complex is completed when larger subunit of ribosome is also placed upon smaller subunit.

(iii) Elongation phase

In this phase ribosomal subunit move along mRNA, Amino acids are brought by tRNA which are joined together to form a protein chain. This is achieved in three steps.The ribosome has three sites “P” site(peptidyl site), “A” site (aminoacyl site) and “E” sie (exit site).

(1) The first aminoacyl complex attaches at “P” site and then the codon of mRNA is exposed at “A” site. Its anticodon bearing aminoacyl tRNA complex binds to it with the help of an enzyme called elongation factor.

(2) Then an enzyme peptidyl transferases from P site, removes the amino acid from tRNA present on P side and binds it to the newly coming amino acid with the help of peptide bond.
(3) Then ribosomal subunits slightly move along mRNA from 5 to 3 direction, so that a new codon is exposed at a site this movement is called translocation. As a result the empty tRNA is reached at "E" site to leave the ribosome, while the other tRNA bearing a chain of amino acid is shifted from "A" site to "P" site and another codon is exposed to "A" site.
These three steps are repeated again and again until the stop codon is reached at "A" site.

(iv)Termination
When ribosome reaches at the stop codon, which do not bind to any tRNA but they are recognized by releasing factor that terminate the process of translation and the polypeptide chain is released from tRNA. The ribosomal subunits separate from mRNA .

Differences in protein synthesis between prokaryotic and eukaryotic

PROKARYOTES	EUKARYOTES
Prokaryotes have ribosomes in cytoplasm so protein synthesis completely takes place in cytoplasm.	Ribosomes are mainly attached with rough endoplasmic reticulum where mostly protein synthesis occurs.
The transcription and translation both occurs in cytoplasm.	The transcription takes place in nucleus while translation occurs in cytoplasm.
Prokaryotic gene do not contain introns.	Gene contain exon and intron regions.
In prokaryotes the promoter regions are TATAAT -10 and TTGACA -35 sequence	The promoter regions are TATA -25 and CAAT -70 sequence
The mRNA is not protected with cap and tail	The mRNA Is protected with the cap of 7 methyl GTP and a Poly A tail.

Regulating gene expression

The process of turning on and off the gene is called regulating gene expression. Each gene is responsible to produce a protein in the form of enzymes for regulating its function , developing its structure and for differentiation, therefore a mechanism is required which controls the production of this protein that is when and how its genes are expressed and when it will be stopped.

Importance of regulating gene expression

The gene expression is regulated to conserve energy and space like the gene expression requires energy therefore the gene only expresses at the time of its requirement and saves energy, secondly the DNA is present in the coiled form so to express a gene only that part of DNA uncoils where required gene is located and saves the space.

In multicellular animals , all the cells contain same genome but only specific genes are expressed in particular type of cell to regulate normal metabolism. If this gene expression is disturbed then it leads to metabolic disorders like cancer etc.

For example liver cells perform breaking of alcohol by producing an enzyme called alcohol dehydrogenase but the red blood cells cannot do it because they keep this gene turned off. Similarly liver cell cannot produce hemoglobin and keep this haemoglobin producing gene turned off.

Methods of regulation of gene expression

There are two possible ways of regulation of gene expression positive regulation and negative regulation.

Positive regulation

When the expression of gene is increased by presence of specific regulatory protein (the activator) is called positive gene regulation.

Negative regulation

When the expression of gene is inhibited by the presence of specific regulatory protein (repressor) is called negative gene regulation.

For example E coli bacteria growing in presence of lactose produces an enzyme which allow lactose to be utilized as source of energy if lactose is absent then the genes required for lactose metabolism are suppressed by a repressor protein.

Role of intron an exon in gene expression

The eukaryotic gene contains introns (non coding sequence) and exons (coding sequence) while prokaryotic gene does not contain intron. During gene expression the introns are copied from DNA into RNA but introns do not participate in translation because before coming out of nucleus the intron regions are eliminated by spliceosome.

Introns play following roles

- Because of intron variety of proteins can be produced from a single gene.
- Introns play an important role in positive regulation of gene expression
- Introns are involved in transcription, initiation and termination process.

Mutation

Any sudden change in the genetic material of an Organism which affects phenotype and changes the specific characters of the organism is called mutation. The agents which cause mutation are called mutagens and the Organism in which mutation occurs is called mutant.

The mutation includes change in sequence of nucleotide in a gene, change in number of chromosomes, change in structure of chromosome etc.

Types of mutation on the basis of importance

A mutation can be negative, positive and neutral mutation.

Negative mutation

The mutation which is harmful for living organism and decreases the fitness of Organism in the environment is called negative mutation. It includes developmental abnormalities such as microcephaly, cleft palate, abnormal number of chromosomes like Down syndrome, Kelinefelter's syndrome, Turner syndrome and hereditary disorders like sickle cell anemia, phenylketonuria etc.

Positive mutations

The mutation which produces better alleles which can adapt to the changing environment is called positive mutation for example polyploidy which produced different varieties within the same species of plants, genes of beak and claws in birds.

Neutral mutation

The type of mutation which does not cause any harmful effect to the organism and inherit from one generation to another generation in a normal pattern are called neutral mutations. For example the land plants and animals are evolved from aquatic animals and plants.

Types of mutation on the basis of mutagenesis

The creation or origin of mutation is called mutagenesis, it includes Spontaneous & Induced mutations.

Spontaneous mutations

The mutations which occur naturally and automatically due to internal or external factors are called spontaneous mutations

Induced mutations

The mutations which are caused artificially for developing new varieties of organisms are called induced mutations.

Types of mutation on the basis of (Location) i.e where they occur and to what extent

These can be of two types

(1) Gene mutation

The mutation which changes structure of gene (change in nucleotides), or position of gene on the chromosome is called gene mutation. It can be of two types point mutation and transposition. Here we will discuss point mutation only.

Point mutation

If one or more nucleotide sequence are changed in a gene is called point mutation. This results in production of non functional protein or a different protein which develops a new character in the organism.

Example: sickle cell anemia, phenylketonuria

Sickle cell anemia

It is a disorder in which the affected individual has homozygous genotype for hemoglobin producing genes which produce defective hemoglobin. As a result the red blood cells of the individual becomes sickle shaped and the hemoglobin is unable to transport oxygen.

Symptoms

The patient feels fatigue, pain, fever, fast heartbeat, breathlessness etc.

Causes

The genetic code at the 6^{th} position for glutamic acid in beta chain of hemoglobin is replaced by a genetic code for valine due to this point mutation defective hemoglobin develops.

Effect of sickle shaped RBC's

The abnormal hemoglobin in red blood cells causes RBC's to a sickle shaped cell and this hemoglobin can not transport oxygen to the body tissues.

Treatment

Blood transfusion, analgesics and bone marrow transplant is the treatment.

Phenylketonuria

The disorder in which a new born individual is unable to break phenyl alanine amino acid into tyrosine amino acid due to lack of phenylalanine hydroxylase enzyme as a result phenyl alanin is converted into phenyl ketone which damages the nervous system. This condition appears in homozygous recessive individuals.

Treatment

Treatment involves a diet that is extremely low in phenylalanine content particularly when the child is growing. Special formula lofenalac is made for infants with phenylketonuria.

(2) Chromosomal mutation

The change in number of chromosomes in an organism or change in structure of chromosome is called chromosomal mutation.

This can be of two types heteroploid and chromosomal aberration

(i) Heteroploidy

The change in number of chromosome due to non disjunction (inability of chromosomes to separate) during meiosis is called heteroploidy. This can be of two types

(a) Polyploidy (euploid)

The gain of one or more complete set of chromosomes is called polyploidy or euploid. This can be triploid (3N) in which one complete extra set of chromosome is present, tetraploid (4N) in which two extra sets of chromosomes are present etc. This type of mutation is responsible to produce different varieties of plants like wheat, rice etc.

(b) Aneuploidy

The type of mutation in which one chromosomes is either increased (trisomy) 2N+1 or One chromosomes is decreased (monosomy) 2N-1.

Ex: Down's syndrome, Kelinefelter's syndrome, Turner's syndrome

Down's syndrome (45+XY) OR (45+XX)

It is a trisomic condition which is characterized by 2N+1. They have an extra copy of 21st autosomal chromosome. This condition can occur in both male and female human beings.

Symptoms

Small size of head, rounded inner corners of eyes instead of pointed, folds of skin at the back of neck, flat nose, single crease in the palm of hand, small ears, short hands with short fingers, long tongue, mentally abnormal, Short life span.

Treatment

There is no treatment but continuous psychological counseling and some medicine required to reduce the aggression of patient.

Kleinfelter's syndrome (44+XXY)

It is also a trisomic disorder 2N+1 which occurs due to increase in Sex chromosome. It is a male disorder.

Symptoms

It is a male disorder having feminine characters, less body hairs and facial hairs, enlarged breasts, long neck, wide hips, curved shoulders and thighs. This male has small tests and is infertile.

Treatment

Testosterone therapy is the only solution but it does not help in reducing infertility.

Turner's syndrome (44+XO)

It is a monosomic disorder 2N-1 in which the affected individual female lack an X chromosome.

Symptoms

Infertile female with short height, webbed neck, low hairline at the back of neck, edema in hands and feet, low IQ level etc.

Treatment

There is no specific treatment, however female(estrogen) hormonal therapy helps in development of secondary sexual characters and reproductive cycle. This treatment can be started when a girl reaches about the age of 12 or 13.

(ii) Chromosomal aberration

The change in structure of chromosome which include deletion, duplication, translocation and inversion.

(a) Deletion

The mutation in which a segment of chromosome breaks and lost as a result the size of chromosome is reduced.

(b) Duplication

The mutation in which a segment of chromosome breaks and rejoins with the homologous chromosome. This results in duplication of genes and the size of chromosome increases.

(c) Translocation

The mutation in which a segment of chromosome breaks and is transferred to a non homologous chromosome is called translocation. This results an increase in the size of chromosome.

(d) Inversion

The type of mutation in which a segment of chromosome breaks at two points and rotate at 180 degree and rejoins the same chromosome is called inversion.

Sources of mutation

The agents which cause mutation are called mutagens which can be physical mutagens, chemical mutagens and biological mutagens.

(i) Physical mutagens

It includes the physical forces which can break any type of bond present in DNA. They include ionization radiations like gamma rays, X-rays, ultraviolet radiations, cosmic waves, ultrasonic waves etc.

(ii) Chemical mutagens

These include the chemicals which can change the shape of DNA. Like some chemicals are similar to the nitrogenous bases and binds with the DNA as a result the structure of DNA changes, some other chemicals like nitrous oxide, mustard gas, photo chemicals or radioactive isotopes may also work as chemical mutagens.

(iii) Biological mutagens

Some microorganisms may also work as mutagens and causes changes in the structure of DNA. They include viruses like HIV, transposons (jumping genes) etc.